A Locality-Sensitive Hashing Framework for Reducing BDD to EDCP

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Motivating question

What structural property of a metric decoding problem enables a quantum reduction from BDD to EDCP?

BDD and EDCP

Definition (BDD)

Let (X, d) be a metric space, $\mathcal{D} \subseteq X$ a discrete subset. Given $t \in X$ and $r < \frac{1}{2} d(\mathcal{D})$ such that $d(t, \mathcal{D}) \leq r$, find the unique $s \in \mathcal{D}$ such that $d(t, s) \leq r$.

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Definition ((Uniform) EDCP)

Let A be a finite Abelian group, $M \geq 2$ an integer. Let $s \in A$ be fixed, $x_i \in A$ arbitrary (unknown). Find s from ℓ states

$$|\psi_{x_i,s}\rangle := \frac{1}{\sqrt{M}} \sum_{j \in [M]} |j\rangle \, |x_i + js\rangle \, .$$

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- DCP: $A = \mathbb{Z}_N, q = N, M = 2$

BKSW reduction sketch

Setup:

- q-ary lattice $q\mathbb{Z}^m \subseteq \Lambda \subseteq \mathbb{Z}^m$ (LWE: $\Lambda = A\mathbb{Z}^n + q\mathbb{Z}^m$)

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1. Prepare $\sum_{j\in[M],y\in\mathbb{Z}_q^n} |j\rangle\,|y\rangle$.
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$$\sum_{j\in[M]} \sum_{x\in\mathcal{D}} |j\rangle\,|x + js\rangle\,|x - je\rangle\,.$$

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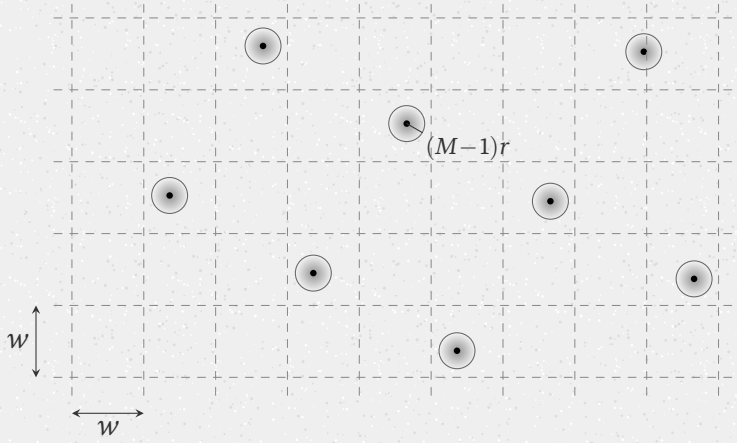
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3. Measure $c = g(x - je)$, where g is a randomly shifted grid of cubes of width w on $\mathbb{R}^m/q\mathbb{Z}^m$, and uncompute the third register:

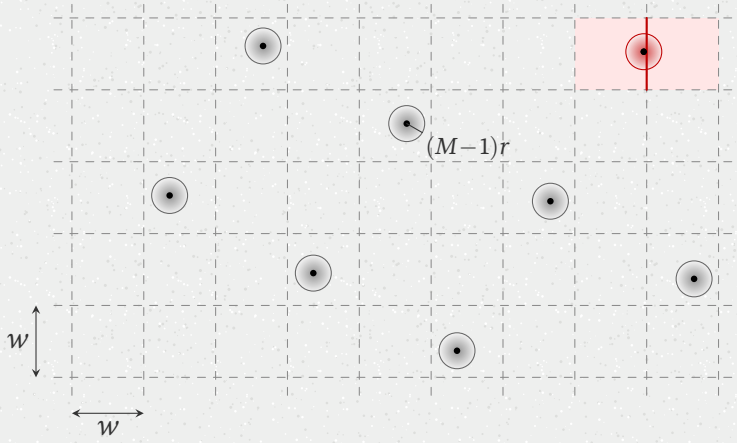
$$\sum_{(j,x)\,:\,g(x-je)=c} |j\rangle\,|x + js\rangle \stackrel{?}{=} \sum_j |j\rangle\,|x + js\rangle\,.$$

Partition and measure



Dots: $x \in \mathcal{D}$. Disks: where $x, x-e, \ldots, x-(M-1)e$ can lie.

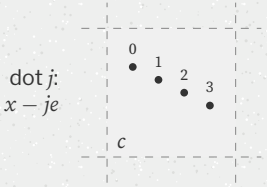
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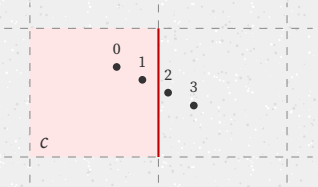
When is the output an EDCP state?

(a) cell c contains the progression



$\frac{1}{\sqrt{M}} \sum_{j \in [M]} |j\rangle |x + js\rangle = |\psi_{x,s}\rangle$ ✓

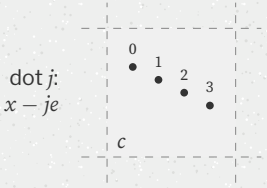
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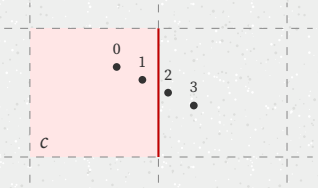
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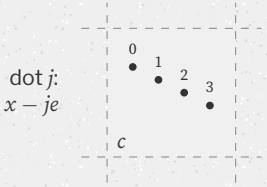


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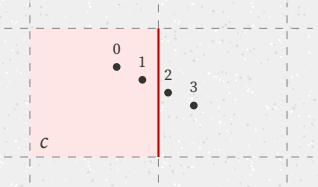
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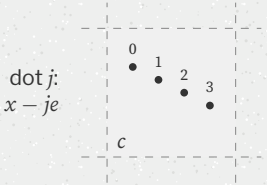


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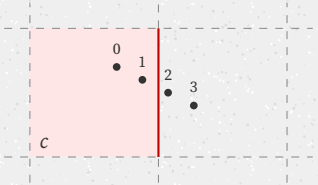
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- All ℓ runs succeed w.p. $\geq \left(1 - \frac{mMr}{w}\right)^{\ell}$, so with $w \approx d(\mathcal{D})/2$ we get $\ell = \Theta\left(\frac{d(\mathcal{D})}{mMr}\right)$ samples.

The abstraction: LSH and convexity

Definition (Locality Sensitive Hash (LSH))

Let (X, d) be a metric space, S a set, $0 \leq r_1 < r_2$. A family of hash functions $\mathcal{F} = \{h : X \to S\}$ is called (r_1, r_2, p_1, p_2)-sensitive if, for any $a, b \in X$

- if $d(a, b) \leq r_1$ then $\Pr_{h \leftarrow U(\mathcal{F})}[h(a) = h(b)] \geq p_1$;
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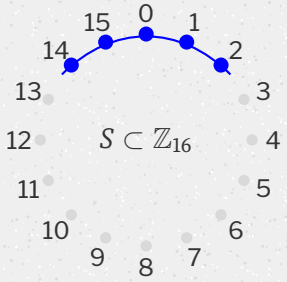
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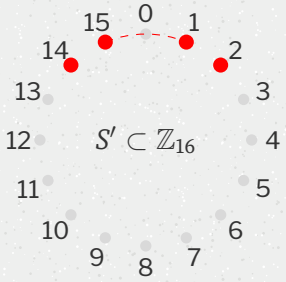
Definition ((r, M)-Convex)

Let G be an additive Abelian group equipped with a weight wt. Let $r \in \mathbb{R}_+$ and $M \geq 2$ be an integer. We say a subset $S \subseteq G$ is (r, M)-*convex* if for every $x, e \in G$ such that $\mathrm{wt}(e) \leq r$, if $x \in S$ and $x + (M-1)e \in S$, we have $x + je \in S$ for all $j \in [M]$.

(r, M)-Convex



(a) $(1, 3)$-Convex Subset



(b) Non-Convex Subset

(a) S is $(1, 3)$-convex as it contains all intermediate points for steps of size 1. (b) The set S' is not $(1, 3)$-convex because, for $x = 15$ and $e = 1$, both endpoints x and $x + 2e = 1$ are in S', but the midpoint $x + e = 0$ is missing.

Main theorem

Theorem (BDD $\to$ EDCP, informal)

Let $\mathcal{D} \leq A$ be finite with minimum distance d, with $|\mathcal{D}\rangle$ efficiently preparable. Let $t = s + e$ be a $\mathrm{BDD}_{\mathcal{D},r}$ instance, and let $\Delta < d/2$ bound the diameter of the progression: $\mathrm{wt}\big((j - j')e\big) \leq \Delta$ for all $j, j' \in [M]$. (Lee: $\Delta = (M-1)r$; Hamming, rank: $\Delta = r$) Let $\mathcal{F}$ be an (r, M)-convex, $(\Delta,\ d - \Delta,\ p_1,\ p_2)$-sensitive LSH family, and

$$p_{\text{iso}} := p_1 - M(|\mathcal{D}| - 1)\, p_2.$$

Then an efficient quantum procedure outputs, with probability at least p_{iso}, a fresh EDCP sample $|\psi_{x,s}\rangle$ with $x \leftarrow \mathcal{D}$ uniform, and some other state otherwise.

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- Failures are not detected, so all ℓ samples are good only w.p. p_{iso}^{ℓ}.
- Hence $\mathrm{BDD}_{\mathcal{D},r}$ reduces to $\mathrm{EDCP}^{\ell}_{\mathcal{D},M}$ in quantum poly time whenever $p_{\text{iso}}^{\ell} \geq 1/\mathrm{poly}$, i.e. for $\ell = O\big(\log n / \log(1/p_{\text{iso}})\big)$.

Case study: Hamming metric

- **LSH** (bit sampling, Indyk–Motwani): $y \in \mathbb{F}_q^n$ ($q = p^N$), sample $i_1, \ldots, i_t \leftarrow [n]$ and set $h(y) = (y_{i_1}, \ldots, y_{i_t}) \in \mathbb{F}_q^t$.

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  With $r_1 = r$ and $r_2 = d - r$: $\quad p_1 = \left(1 - \frac{r}{n}\right)^t, \quad p_2 = \left(1 - \frac{d-r}{n}\right)^t$.

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Reality check

Even one sample needs $r = O(\log n)$, where ISD already decodes in polynomial time. LPN at noise rate $\eta = \Theta(\log^2 n/n)$: $\ell_H = \Theta(1/\log n)$.

Why it breaks: what a bucket looks like

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Bit sampling (Hamming metric)

- Bucket: all y agreeing on the t sampled coordinates.

Why it breaks: what a bucket looks like

Grid (∞-Lee metric, BKSW)

- Bucket: a cube of width w.
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- All t samples must miss the error: $p_{\text{iso}} \approx \left(1 - \frac{r}{n}\right)^{\Theta(n)} = e^{-\Theta(r)}$.

Summary and open problems

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- Is the loss inherent to isolation, or only to our hash families? (Bit sampling is optimal for nearest-neighbor search, not necessarily here.)
- Structure-aware hashing for quasi-cyclic, module or rank-metric codes.
- The converse, EDCP $\to$ decoding, as BKSW prove for LWE.

Thank you

Questions?

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